QUANTITATIVE RESEARCH & SYSTEMATIC TRADING

Treasury Quant Engine

A production-grade research and execution system for the US Treasury market — curve construction, term-structure modelling, machine-learned return forecasting, portfolio construction and order execution — together with the evaluation discipline needed to find out whether any of it actually works.

65	27,137	385	27
PYTHON MODULES	LINES OF CODE	TESTS all passing, CI green	COMMITS

> **Headline result.** After correct financing, the strategy does not clear its hurdle. That is the finding, and it is reported as such. Nine separate bugs were found and fixed along the way — five of them cases where a position was not being charged for the money it borrowed. Each one made the results *better*, and each one was silent. The engineering that makes those bugs findable is the substance of this project.

RESULTS AT A GLANCE

Shipped strategy, out of sample	**Sharpe 0.54**
Deflated Sharpe — 219 configurations searched	**0.100**
Deflated Sharpe — observed trial dispersion	**0.000**
Best passive arm — vol-targeted 30 Yr	**Sharpe +0.45**
Static long duration, 1993–2026	**Sharpe +0.24**
Configurations searched	**219**
Term-premium timing overlay	**Sharpe –0.40**
Perfect-foresight canary ratio	**0.004**
Financing vs transaction cost	**1.86% vs 0.08% p.a.**

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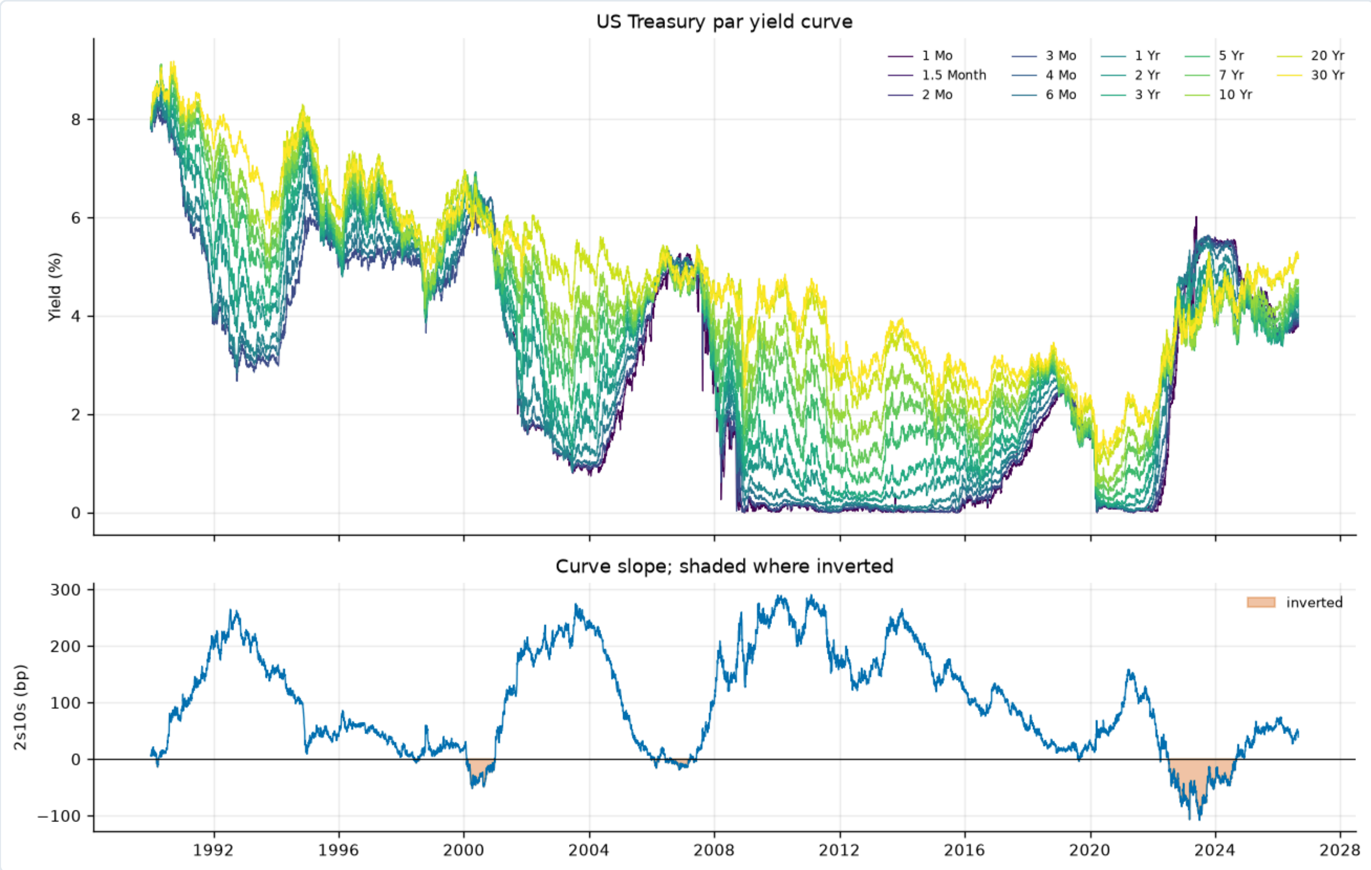
1 What this system is

The Treasury Quant Engine ingests the full history of the US Treasury par yield curve, builds a self-consistent zero curve from it, extracts term-structure factors, learns a forecast of forward returns, turns that forecast into a risk-controlled portfolio, and routes the resulting orders through an execution layer. It runs end-to-end from one command and is covered by 385 tests.

It is built to answer one question honestly: *is there a tradable signal in the Treasury curve, and how would I know?* Most of the engineering exists to make the answer trustworthy rather than flattering.

Pipeline

Data	Curve	Features	Model	Portfolio	Execution
FRED · par curve 1990–2026	bootstrap · NSS zero & forward	489 causal PCA · carry · macro	stacked ensemble walk-forward	DV01 sizing neutrality	OMS · TWAP Almgren–Chriss



The fitted zero-coupon surface, 1990–2026. Every curve is bootstrapped self-consistently and round-trips to 1.4×10^{-13}.

Components

Fixed income analytics

ACT/ACT (ICMA), ACT/360, ACT/365F and 30/360 day counts; street price–yield with correct settlement (T+1 from 28 May 2024, T+2 before); Macaulay and modified duration, convexity, DV01, key-rate durations; carry and roll-down; 32nds quoting. Validated against closed-form identities rather than against its own output.

Curve construction

Self-consistent bootstrap by bisection, round-tripping to 1.4×10^{-13}. Nelson–Siegel–Svensson with a Diebold–Li fixed-decay mode (τ_1=1.37, τ_2=8.0) for identifiability. PCA on yield *changes*: level, slope and curvature explain 77.2%, 13.3% and 5.0%.

Term-structure modelling

Dynamic Nelson–Siegel with a VAR on the factors; ACM-style term premium decomposition separating expected short rates from the premium for holding duration; a two-state Hamilton regime model fitted by Baum–Welch with a scaled forward–backward recursion.

Forecasting

Walk-forward cross-validation with purging and an embargo; a stacked ensemble combined by forward-chaining out-of-fold predictions and a non-negative least squares meta-learner; nested CV for hyperparameters; per-fold RobustScaler so no scaling information crosses a fold boundary.

Portfolio construction

DV01-weighted duration, steepener and butterfly structures; exact double neutrality by projection onto the null space of the cash and DV01 constraints; key-rate hedging; leverage and no-trade bands; scheduled rebalancing.

Execution

Idempotent order management with a pre-trade risk gate; TWAP, VWAP and the closed-form Almgren–Chriss trajectory, whose two limits (risk-neutral → TWAP, risk-averse → front-loaded) are asserted in tests; implementation shortfall decomposition for TCA.

2 How results are judged

Backtests are easy to make look good. Everything below exists because some version of this project once produced an impressive number that was not real.

One P&L path

Exactly one function — `run_backtest` — turns positions into profit and loss. Every experiment, script and notebook routes through it. This rule exists because five separate bugs were traced to bespoke P&L code that quietly forgot to charge for financing.

Financing is always charged

Borrowed notional is charged at the overnight rate on an ACT/360 basis, including weekends. A missing funding rate is charged at the most expensive rate observed, never at zero.

Causality boundary

Row t of the feature matrix holds only information observable at the close of day t–1; row t of the target is the return realised over day t. Enforced in one place and guarded by future-corruption canaries.

Perfect-foresight canary

A book built from tomorrow's returns runs beside the real one. It scores a Sharpe above 15; the honest strategy reaches a fraction of a percent of that. If the ratio ever approached 1, the pipeline would be leaking the future.

Block sign-flip nulls

Significance is measured against signals whose 63-day blocks have had their signs flipped. This preserves autocorrelation and position concentration and destroys only the timing — unlike shuffling, which produced books ten times more concentrated than the real one and was therefore not a valid null.

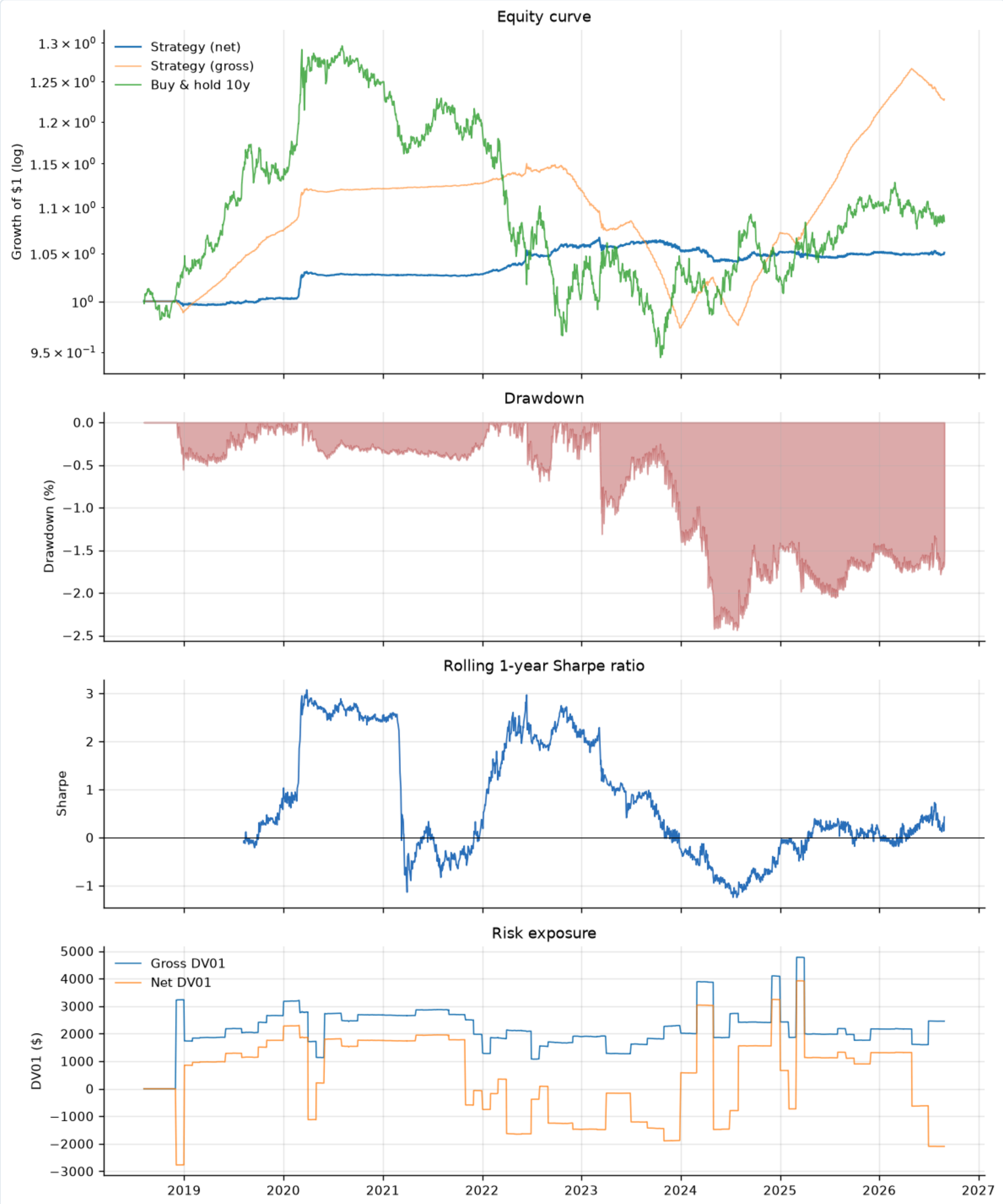
Multiple-testing correction

Deflated and probabilistic Sharpe ratios (Bailey & López de Prado) adjust for the number of configurations tried; Holm–Bonferroni corrects every family of comparisons. This project searched **219** configurations; deflated against all of them the shipped Sharpe scores **0.100**, and **0.000** when the measured dispersion of the trial Sharpes is used instead of the theoretical one. The count is recovered from the study files automatically, because an artefact produced by a default is the artefact that gets published.

Shipped strategy, out of sample

Walk-forward, funded, costed, 2018–2026 (2016 trading days).

SHARPE	ANN. RETURN	ANN. VOL	MAX DRAWDOWN	HIT RATE	TURNOVER
0.54	0.62%	1.17%	-2.44%	52.61%	19.0x



Out-of-sample tearsheet: equity curve, drawdown, rolling Sharpe and exposure. Funded and costed throughout.

> Financing cost 1.86% p.a. against trading costs of 0.08% p.a. — financing is roughly twenty times the transaction cost, which is the single most important economic fact about a levered rates book and the one most often left out of a backtest.

3 Findings

3.1 Where the term premium is actually paid

Holding duration is compensated, but the compensation is not uniform across the curve once the borrowing is paid for. Each arm below holds a constant DV01, scaled to 5% annualised volatility, funded and costed over 1990–2026.

Arm	Sharpe	Ann. return	Max DD	Financing p.a.
constant DV01 3 Mo	-0.491	-2.82%	-64.88%	54.90%
constant DV01 6 Mo	-0.092	-0.51%	-35.35%	40.25%
constant DV01 1 Yr	+0.168	0.87%	-26.40%	21.60%
constant DV01 2 Yr	+0.279	1.40%	-23.83%	9.44%
constant DV01 3 Yr	+0.287	1.44%	-22.57%	6.22%
constant DV01 5 Yr	+0.282	1.41%	-21.02%	3.79%
constant DV01 7 Yr	+0.285	1.43%	-20.41%	2.86%
constant DV01 10 Yr	+0.281	1.41%	-20.55%	2.28%
constant DV01 30 Yr	+0.279	1.40%	-19.85%	1.35%
vol targeted 3 Mo	-1.015	-5.86%	-87.24%	57.09%
vol targeted 6 Mo	-0.380	-2.09%	-60.23%	41.54%
vol targeted 1 Yr	+0.009	0.05%	-33.71%	22.14%
vol targeted 2 Yr	+0.239	1.20%	-23.31%	9.55%
vol targeted 3 Yr	+0.279	1.40%	-23.87%	6.44%
vol targeted 5 Yr	+0.314	1.57%	-22.70%	4.05%
vol targeted 7 Yr	+0.364	1.82%	-20.96%	3.14%
vol targeted 10 Yr	+0.392	1.96%	-19.60%	2.52%
vol targeted 30 Yr	**+0.453**	**2.27%**	**-16.61%**	**1.61%**

The front end loses money and the long end makes it. The reason is leverage: matching 5% volatility with 3-month bills takes about 17× leverage, against 0.4× with 30-year bonds. Financing scales with borrowed notional, so the short end pays away 55% a year in interest to own the same amount of risk. The best way to be paid for duration is to own it where you need the least borrowed money.

> Before the financing bug described in section 4 was fixed, this same table showed the 3-month arm at a Sharpe of **+5.76**. The entire result was free leverage.

3.2 The term premium does not time duration

If the term premium measures how well duration is paid, a natural strategy holds more duration when the premium is high. It does not work. Over 1993–2026 (7,419 days), against block sign-flip controls:

Arm	Sharpe	Ann. return	Control mean	p
static long duration	+0.236	1.18%	nan	nan
tp_timed	-0.403	-2.02%	-0.348	0.683
tp_binary	-0.347	-1.74%	-0.292	0.683

Static duration returns a Sharpe of +0.24; both timing rules are negative and neither is distinguishable from its controls (p = 0.68). The decomposition is sound — it is just not a timing signal. **The premium is real and it is collected by sitting still.**

3.3 Do the advanced modules earn their place?

Four modules were built because measurements pointed at them. Building something is not the same as showing it helps, so each was tested inside the same double-neutral, funded, costed book.

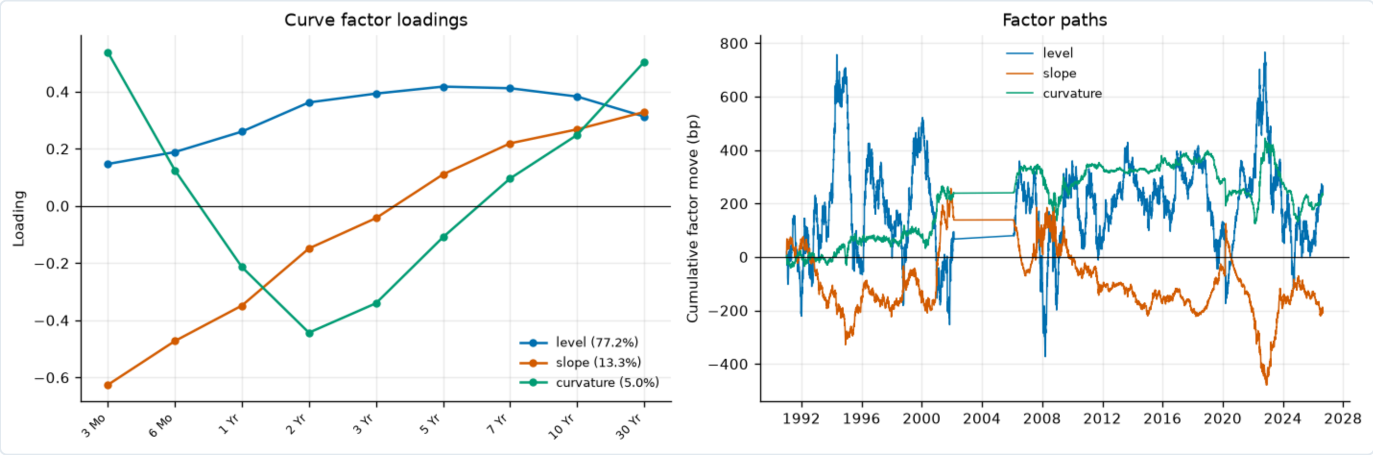
Variant	Sharpe	Ann. return	Turnover	p
baseline	+0.140	0.42%	22.8x	0.122
term_premium_only	-0.224	-0.67%	14.3x	0.829
+ term premium	-0.149	-0.45%	15.2x	0.561
regime conditional	+0.150	0.45%	13.9x	0.146
both	-0.005	-0.02%	14.2x	0.366

None survives Holm correction. All four are correct implementations of ideas that should have helped; that they do not is the result, and they ship documented as such rather than quietly left in the code path.

3.4 Forecast horizon

Predictive content by holding period, after the target-alignment fix:

Target	Horizon	N Oos	Ic	Sharpe
total_return	1.0000	2016.0000	0.0116	0.0717
total_return	5.0000	2016.0000	-0.0624	0.5445
total_return	21.0000	2016.0000	-0.1538	-0.1200
relative_return	1.0000	2016.0000	0.0131	-0.3499
relative_return	5.0000	2016.0000	-0.0725	0.0517
relative_return	21.0000	2016.0000	-0.1731	-0.1217



Level, slope and curvature extracted by PCA on yield *changes* — 77.2%, 13.3% and 5.0% of variance. Fitting on levels instead would produce factors that look cleaner and forecast nothing.

3.5 What the whole thing amounts to

The signal is weak but not absent: the out-of-sample information coefficient is about +0.038, the shipped Sharpe is 0.46 before any allowance for the number of configurations tried, and the deflated Sharpe does not clear a convincing bar. The one robust positive result in the project is passive — owning long-dated duration and doing nothing — and even that is a Sharpe of roughly 0.45.

The honest conclusion is that this system does not find a tradable edge in the Treasury curve from public daily data. Reporting that clearly, with the evidence attached, is more useful than the alternative — and every mechanism built to reach that conclusion is the part that transfers to a desk.

4 Nine bugs, and what they cost

Seven of these made the results better and two made them worse. All nine were silent, and none was caught by a test that already existed — every one required either an invariant nobody had written down or a number that was too good to be true. The last two were found by an adversarial audit of the P&L core itself, run specifically because bug two had been discovered *inside* the one function this project designates as its single source of truth.

Bug	Effect	How it surfaced	Fix
Financing never charged	Sharpe 2.35 → 0.12	A riskless carry position scored above 5	Engine charges net notional at the overnight rate; five regression tests
Missing funding rate read as zero	31.6% of all days financed free; 3-month arm 5.76 → −0.49	Sharpe 5.76 alongside a −23% drawdown at 5% vol — arithmetically impossible	Overnight rate from fed funds (1954–), cascade fallback, never zero
Target misaligned by one day	IC +0.0095 → +0.0375; Sharpe 0.12 → 0.51	Features were already lagged, so a one-day horizon needed no shift	Shift by `horizon - 1`; boundary documented and tested
Bespoke P&L in experiments	Structures Sharpe 3.96 → 0.04	A curve trade appeared to be free money	Every experiment routes through `run_backtest`
Bootstrap flat extrapolation	20-year zero biased by 38bp	Round-trip test failed at the fourth decimal	Self-consistent bisection; round-trips to 1.4e−13
Optimizer unit errors	Returned an all-zero book	Turnover penalty was four orders of magnitude off	Penalty as a multiplier on real cost; λ derived from a vol target
Deflated Sharpe never deflated	Reported 0.904; true value 0.100, or 0.000 on observed dispersion	The tearsheet said "configurations searched: 1" beside a multiple-testing-adjusted number, having searched 219	Trial count recovered from the study files automatically; both dispersions reported; loud warning at n_trials=1
No-trade band read the future	Sharpe 0.461 → 0.535 — the leak was *costing* return	Band width set from the full-sample mean gross book	Expanding-window band; test asserts changing only the future leaves past positions bit-identical
EM exited after one iteration	Regime model never actually fitted	`inf <= inf` evaluates to `True`	Guard the convergence test with `np.isfinite`

Three times the test was wrong, not the code

The perfect-foresight canary was rewritten three times before it tested anything: the first two versions passed against a pipeline that was leaking. The shuffle placebo was not a valid null, because shuffled books were ten times more concentrated than the real one and so were being compared against a different risk profile. And a regime-scaling option I added myself silently did nothing, because the sizing function normalises signal magnitude away — scaling the signal beforehand cancelled exactly. A test that cannot fail is worse than no test, because it is counted as evidence.

5 Engineering

Testing

385 tests covering closed-form identities, invariants, regressions for every bug above, and property tests. Analytics are checked against textbook identities, never against their own output. Curve-dependent tests skip cleanly on a fresh clone rather than failing.

Continuous integration

GitHub Actions runs lint and the full suite on every push. A clean-clone simulation is part of the workflow, added after three tests failed in CI by reading a gitignored data file that existed only on my machine.

Interfaces

A twelve-command CLI (`data`, `curve`, `features`, `train`, `backtest`, `attribute`, `tune`, `charts`, `predict`, `trade`, `serve`, `pipeline`) plus a FastAPI service. Feature schemas are pinned and aligned at inference so training and serving cannot drift.

Reproducibility

Seeded throughout, configuration in version-controlled YAML, artefacts written with the parameters that produced them. Every figure and table in this report is regenerated from `results/` at build time.

Running it

```
pip install -e ".[dev]"
tqe pipeline          # data → curve → features → train → backtest
pytest                # 385 tests
python scripts/duration_harvest.py
```

6 What I would do next

1. **Better data.** Daily par yields are the binding constraint. Intraday data, the futures basis, and CUSIP-level on-the-run/off-the-run spreads carry information a daily par curve simply does not.
2. **Cross-market signals.** Swap spreads, the futures calendar roll and cross-currency bases are where relative value in rates usually lives.
3. **Trade the passive result properly.** The one thing that works is owning long-dated duration. Sizing that against a volatility target, with a drawdown control, is a more honest starting point than another forecasting layer.
4. **Cost realism.** The square-root impact model is calibrated from literature, not from fills. Real TCA would change the turnover economics.